Knowledge Distillation:

Transferring Intelligence from Complex Models to Efficient Systems.

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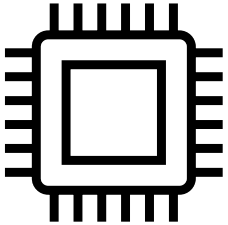
Hardware Constraints



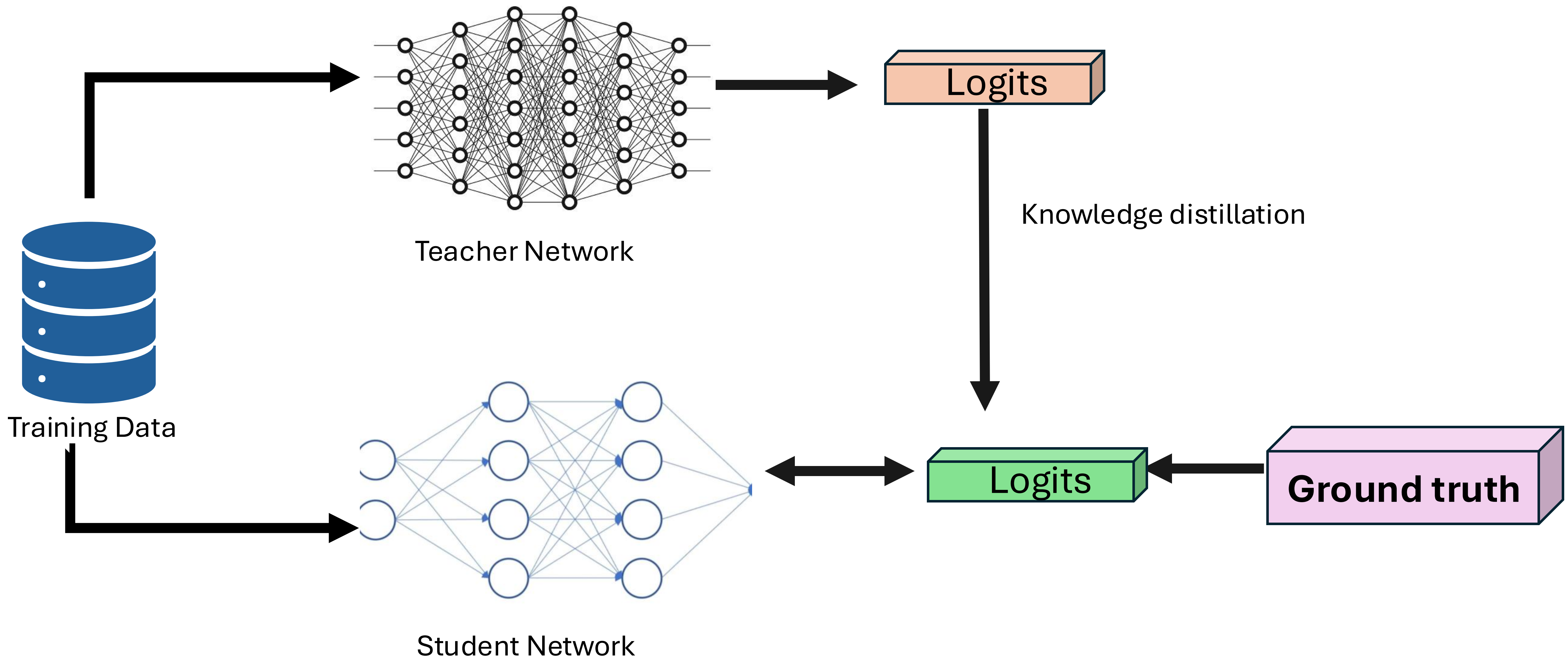
	Cloud AI	**Edge AI**
Computation (fp32)	19.5 TFLOPs (trillion)	MFLOPs (million)
Memory	80 GB	256 kb
Neural Network	ResNet Vision Transformer Med-SAM	MobileNet MCUNet ...

Question and possible solutions

How to train tiny models on large datasets with the help of large model?

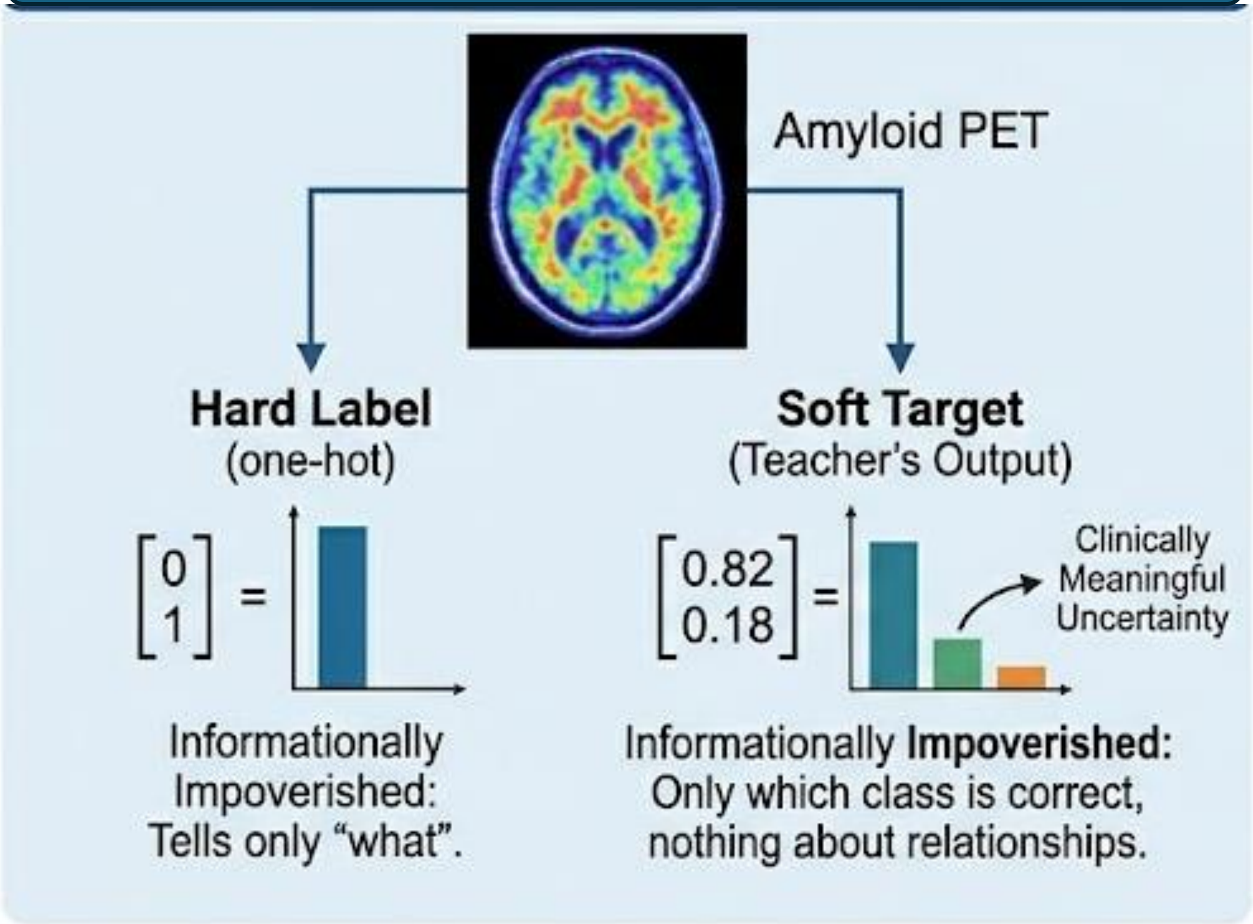
Model Compression:	Network Pruning Quantization
Efficient Architecture:	Squeeze Net Mobile Net
Neural Architecture Search:	Searching Efficient architecture
Knowledge Distillation:	What we want to learn now.

What is knowledge distillation?

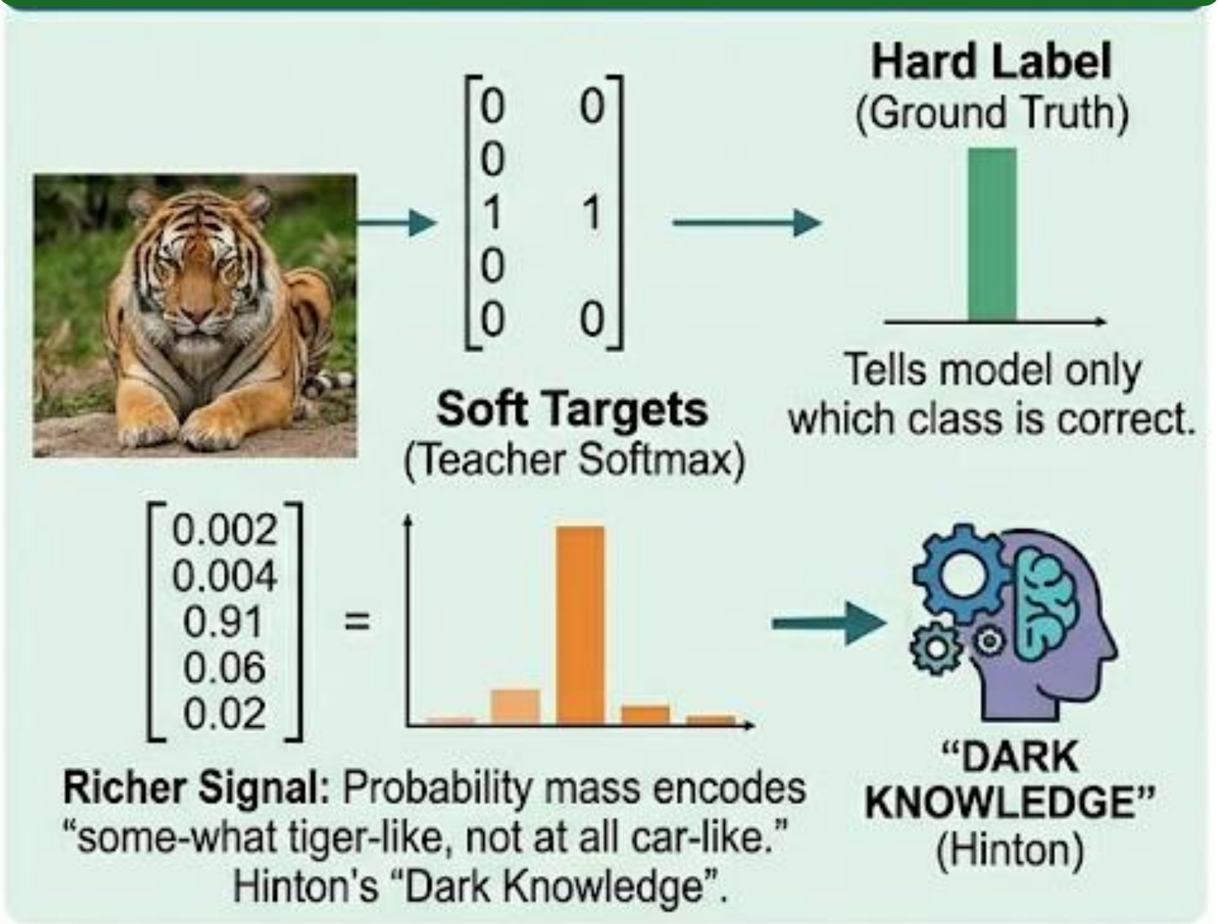


Soft Target and Hard Target.

Amyloid PET Imaging

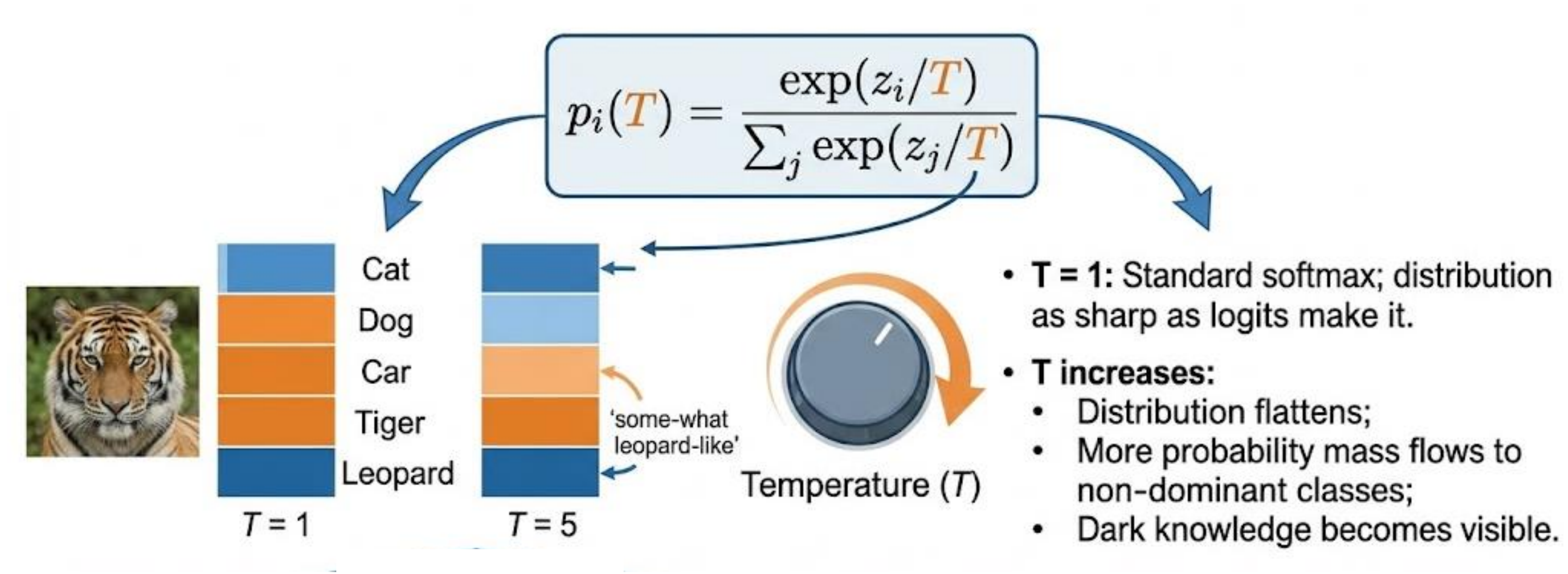


General Case Multi- Class Classification



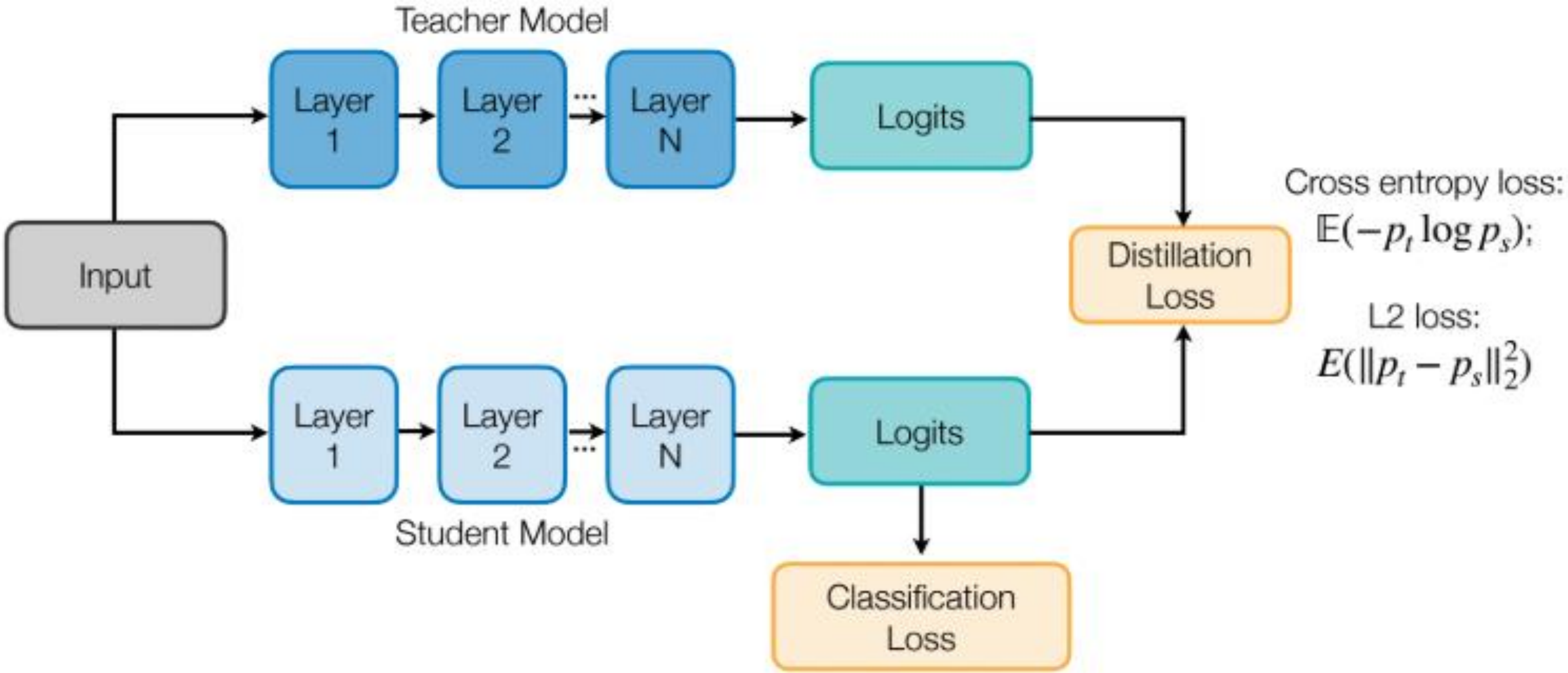
Empirically, students trained with soft targets converge faster and generalize better than those trained from scratch on hard labels, even when the student has seen the same training data.

Temperature Scaling.



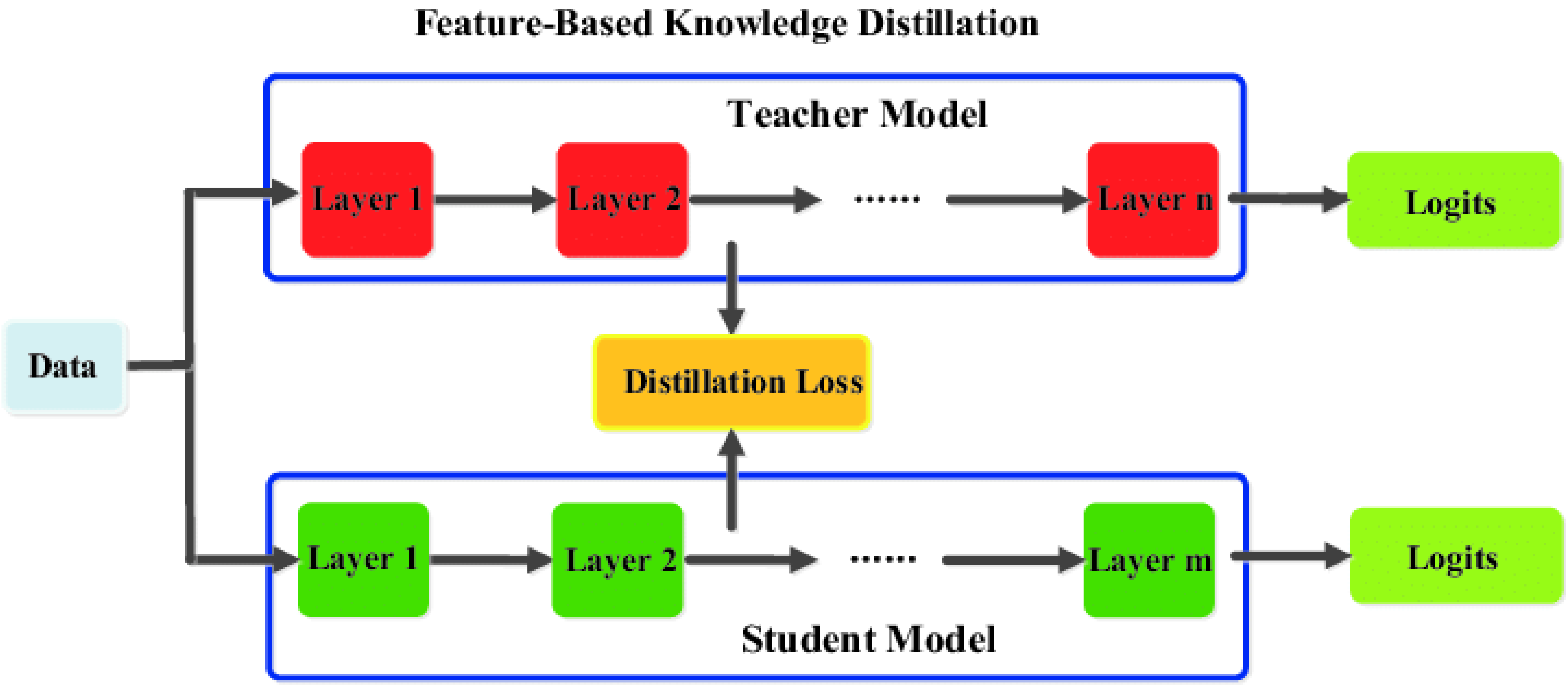
Response Based Distillation

Learning From Soft outputs of the teacher model

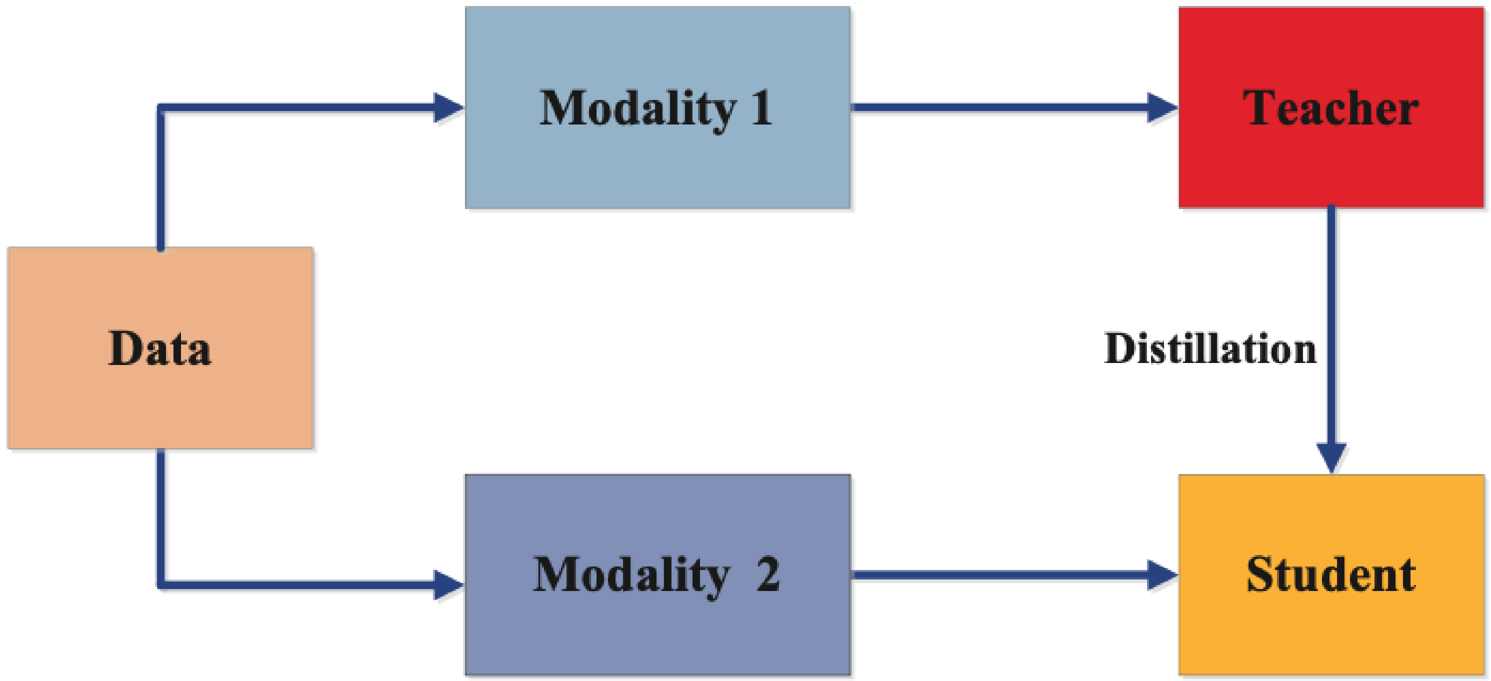


Feature Based Distillation

Learning from Intermediate feature representations



Cross Modal Distillation

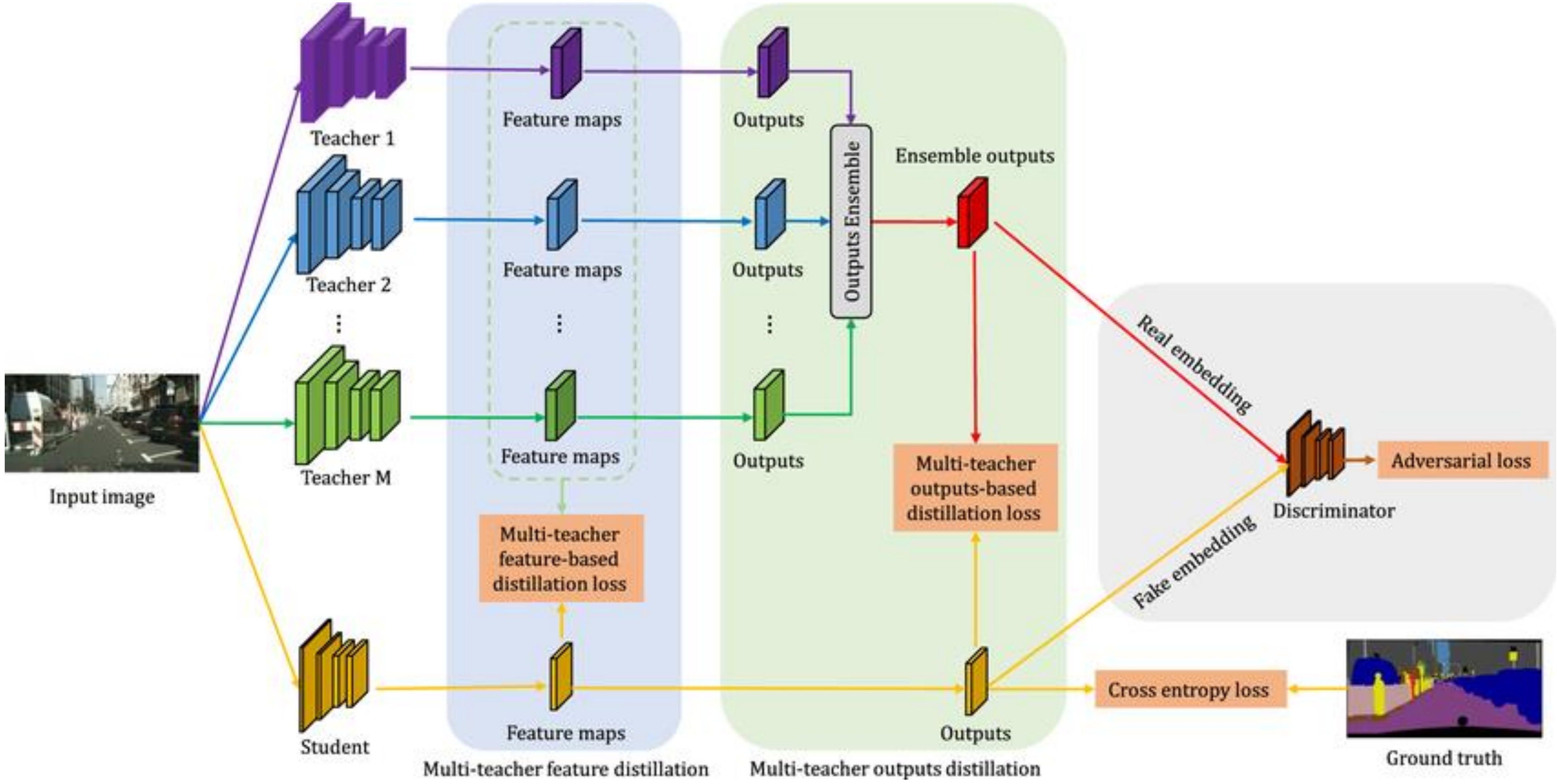


Useful in Medical Imaging where multiple modalities can help in making more informed diagnosis.

MRI and PET images for Alzheimer's disease

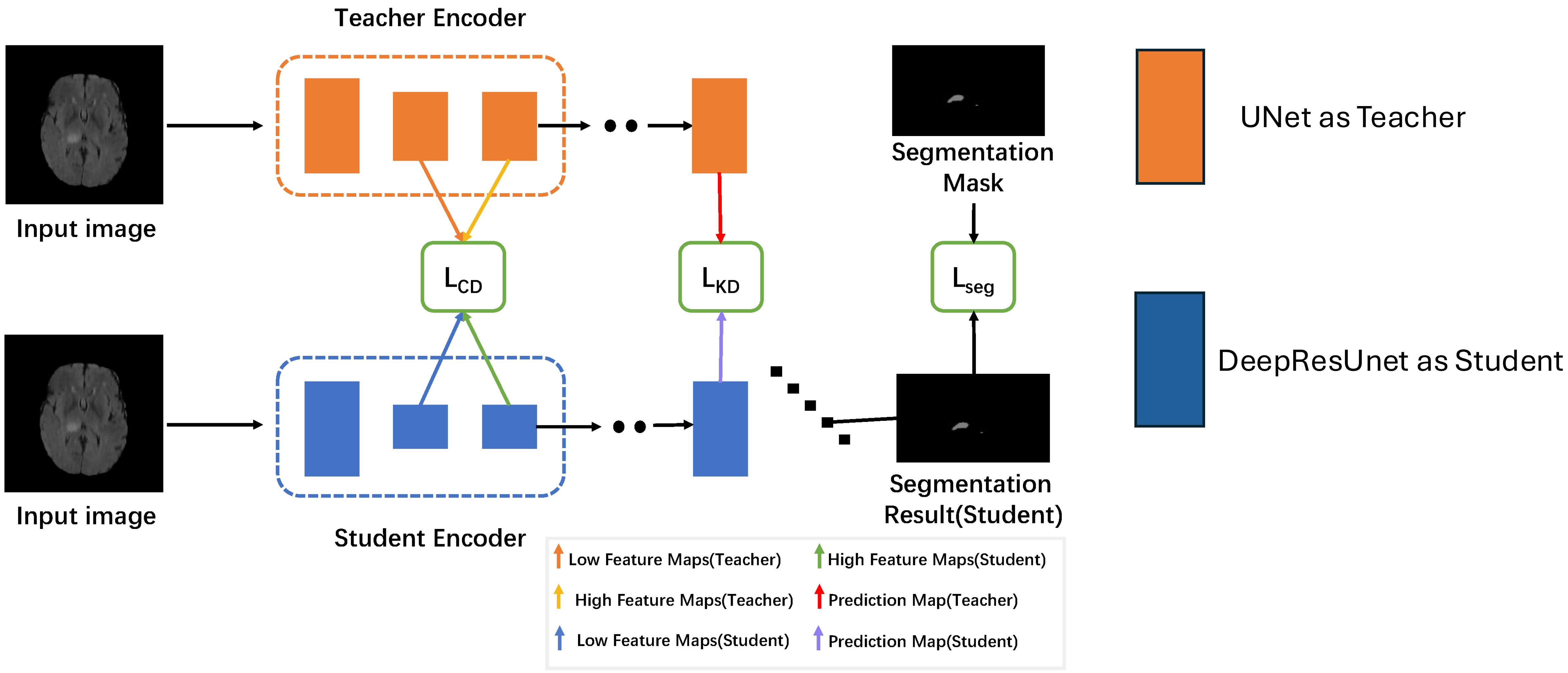
Mig Gu et al., A Cross-Modal Mutual Knowledge Distillation Framework for Alzheimer's Disease Diagnosis: Addressing Incomplete Modalities, 2024

Ensemble /Multi-teacher Distillation



Wang, C., Zhong, J., Dai, Q. et al. MTED: multiple teachers ensemble distillation for compact semantic segmentation

Example: Knowledge Distillation in Brain Tumor Segmentation.



Yuan Qi et al., Efficient Knowledge Distillation for Brain Tumor Segmentation, 2022

Training Procedures

Step	Description
1. Train Teacher	On real labels
2. Build Student	Lightweight architecture
3. Distillation Loss	Combine teacher signals + ground truth
4. Train Student	Minimize distillation + supervised loss
5. Evaluate	Accuracy, speed, size

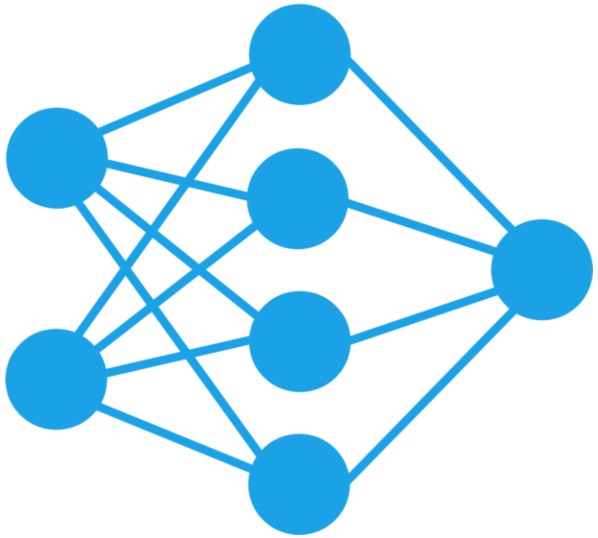
Step 1: Train the Teacher Model

- Use Large models like ViT or Ensemble, or Med-SAM, nnUNet, MedNext, …
- Train the model using ground-truth labels with standard loss function
- Save the model weights after satisfactory performa nce

```
teacher = TeacherModel()
optimizer = torch.optim.Adam(teacher.parameters())
criterion = nn.CrossEntropyLoss()

# Train on labeled dataset
for images, labels in dataloader:
    outputs = teacher(images)
    loss = criterion(outputs, labels)
    # Backprop, step, etc.
```

Step 2: Define Student Model



- Define a smaller model with fewer parameters.
  - Tip: Use a light-weight model.
  - Eg (UNet for segmentation, MobileNet for Classification,...
- It should be capable of mimicking teachers output or intermediate features

Step 3: Define the Distillation loss

KL Divergence Loss (Logits level)

```
loss = (1 - alpha) * CE_loss(student_logits, labels) +
       alpha * KLDivLoss(student_logits / T, teacher_logits / T) * T**2
```

Contrastive Loss (Intermediate level)

```
loss = contrastive_loss(student_feature, teacher_feature)
```

Step 4: Train the Student Model

- Using the distillation loss in Step 3 to train the model defined in step 2.

```
teacher.eval()  # Freeze the teacher model
student.train()

for images, labels in dataloader:
    with torch.no_grad():
        teacher_outputs = teacher(images)

    student_outputs = student(images)

    # Example: logits-level distillation with cross-entropy
    hard_loss = F.cross_entropy(student_outputs, labels)
    soft_loss = F.kl_div(
        F.log_softmax(student_outputs / T, dim=1),
        F.softmax(teacher_outputs / T, dim=1),
        reduction='batchmean'
    ) * T**2

    loss = alpha * soft_loss + (1 - alpha) * hard_loss
    # Backprop, step, etc.
```

Step 5: Evaluate the Student Model

After training:

Evaluate on validation/test sets.

Compare accuracy, inference speed, and model size vs. the teacher.

Discussion points

Can a student model ever outperform its teacher? Why or why not?

What are some ways to evaluate whether distillation was successful?

What could go wrong if you forget to eval() the teacher model during distillation?

If your student model is not learning well, what are some things you might check or adjust?

Summary

- Knowledge distillation helps to reduce memory and computation footprints.
- All SOTA large language models (DeepSeek, ChatGPT-o3…) releases a distilled version but its still computationally heavy
- In medical imaging, distill foundation models can be used in low-resource settings.
- Still an active area of research in low powered medical imaging